

Super-Resolution Reconstruction of Remote Sensing Images Based on GAN

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Abstract—In recent years, the methods of super-resolution image reconstruction that based on deep learning have become a hot topic in research of computer vision. The methods of super-resolution image reconstruction that based on the Generative Adversarial Network (GAN) are not controlled in network generation, the models are easy to collapse, the generalization ability is undesirable, and the time complexity degree is too high. To fill these gaps, we propose a super-resolution image reconstruction method based on the GAN of encoding and decoding, which improves the quality of image reconstruction. First of all, our approach uses a design network with regularized structure to avoid model collapse. Then we build a generation network structure that based on encoding and decoding to suppress the uncontrollable defects of GAN network generated images. Finally, in the last layer of the generator, $N \times N$ convolutional feature layer is included to replace the Softmax layer, which speeds up the training of the model. The experimental results show that the super-resolution remote sensing image reconstructed by the proposed method has higher reconstruction quality and better generalization ability in the DOTA training data sets. At the same time, the image reconstruction process can take much less time.

Keywords—GAN, Coding and Decoding, Remote Sensing Images, Super Resolution Reconstruction

I. INTRODUCTION

Nowadays, remote sensing images have been widely used in surveying and mapping, environment, ecology, tourism and other fields. Remote sensing images with ultra-high-definition resolution can more clearly reflect the geographic information of the designated area. Limited by the hardware cost of ultra-high-resolution remote sensing satellites and the bandwidth remote sensing image transmission, the low-resolution remote sensing image super-resolution reconstruction technology is widely used in such field.

The concept of super-resolution reconstruction was first proposed by Harris et al [1]. Its main idea is to use low-quality and low-resolution images to generate a single high-quality and high-resolution image. However, the current super-resolution reconstruction is roughly divided into reconstruction-based methods and learning-based methods.

The method based on reconstruction is to use statistical balanced sampling and unbalanced sampling theorems, such as Bayesian analysis, maximum posterior probability and aliasing cancellation. These types of methods usually require image registration in advance, and its operation is complicated, versatility is undesirable, and the time complexity degree is too high. The method based on learning

is to first use a large number of high-resolution images to create a training library, and then use different compression algorithms for these high-resolution images to obtain low-resolution images. Finally, the mapping relationship between low-resolution images and high-resolution images is found through the designed algorithm.

At present, the research on super-resolution reconstruction of remote sensing images is relatively mature, but there is still a big gap from practicality, which is mainly reflected in the following aspects:

- The success of super-resolution enhancement of remote sensing images depends on an accurate degradation model that meets the characteristics of the actual imaging system and imaging conditions, but it is very difficult to obtain a degradation model that can meet the actual imaging process.
- Efficiency and robustness issues. The computational complexity of the current resolution algorithm is high because various assumptions are added to the algorithm, such as illumination, contrast, salt and pepper noise, and so on.
- Lack of effective and objective evaluation criteria. At present, for super-resolution reconstruction of remote sensing image using PSNR stage, MSE. In most cases is to rely on subjective assessment, which is not able to form an objective evaluation criteria.

Dong et al [2]. built a three-layer convolutional neural network to simulate the process of extracting the mapping between low-resolution and high-resolution images in sparse representation. However, because the structure designed by Dong is too simple, and the effect of reconstruction is only based on the training data pair, the overall generalization ability of the model is relatively poor.

Due to the single pair of low resolution and high-resolution image data in the scheme proposed by Dong, in 2017, Goodfellow et al [3]. Uses the latest Generative Adversarial Networks (GAN) [4-8] generators to generate random low-resolution images, so that the discriminator network can determine whether these randomly generated low-resolution images meet the reconstruction requirements. However, the GAN network itself has defects and limitations. The discriminator of the model always converges faster than the generator, and most of the low-resolution images generated are very similar, resulting in the overall generalization ability of the model is still very poor.

In view of the shortcomings of the current remote sensing image super-resolution reconstruction in terms of computational efficiency and model generalization performance, a new remote sensing image super-resolution reconstruction method is proposed. Our contributions can be summarized as follows:

- Based on Goodfellow's design principle, regularization is used to limit the convergence speed of the discriminator, and the generator generates the network with priority convergence, which is used to improve the effect of the degradation model of high-resolution image.
- An instance regularization is added to the generator network to restrict and constrain the color space and style of a single image after reconstruction.
- In terms of data set setting, in addition to relying on the generator of GAN to generate some low-resolution images, we also use the linear difference method and the nearest neighbor difference method to obtain the low-resolution data of the original HD image respectively to alleviate the model collapse caused by GAN generator.
- The discriminator network uses the grid probability mean to replace the probability value of Softmax, which can adjust the size of the grid according to the demand, so as to alleviate the computational efficiency of the model.

II. RELATED WORK

In this section, we firstly summarize related work on the basic principle, structure and function of GAN, and then its shortcomings are analyzed and clarified. Finally, we came up with our own solutions.

A. GAN

The main idea of Generative Adversarial Networks (GAN) is to design generator network and discriminator network to compete with each other and finally achieve Nash equilibrium. The network structure principle is shown in Figure 1. The generating network is used to generate false data, and the discriminator network is used to distinguish the true and false data of the generating network. The output range of the discriminator network is a scalar of $[0, 1]$, which represents the probability that the discriminating data is true.

The model of the discriminator network in Fig.1 continuously learns and optimizes its own parameters so that the generated data can pass through the discriminator. When the generator network and the discriminator network reach Nash equilibrium, they cannot improve their parameters through learning. At this time, the discriminator network determines that the generator network outputs real data, and the probability is 0.5:

$$\min_G \max_D V(D, G) = E_{x \sim P_{data}(x)} [\log D(x)] + E_{z \sim P_z(z)} [\log(1 - D(G(z)))] \quad (1)$$

Equation (1) is the optimization function of GAN, which can make the network reach the equilibrium point under

normal circumstances. However, in the actual training, because the discriminator of the network will be designed too powerful, the generator cannot learn more image features.

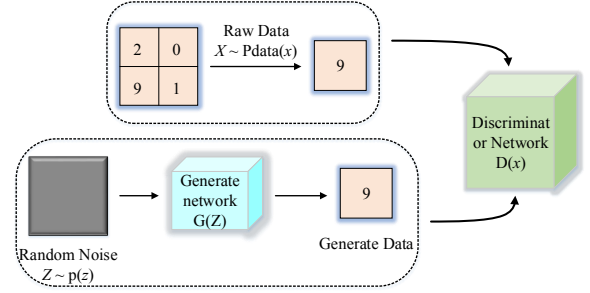


Fig. 1. Generative adversarial nets schematic diagram.

B. Improvements

In order to overcome the instability in the training models of GAN, it is necessary to limit the training conditions of the original network. For the image task, a lot of information is shared between the input and output of function G , such as the edge information of the image. Therefore, in order to ensure the similarity between the input image and the output image, smooth $L1$ regularization loss is added. As shown in Equation (2), defining (x, y) is a real image pair, and the image generated by G is $G(x)$, then it should be close to the real image. Different from the traditional $L1$ regularization, we use a smoother $L2$ regularization when x is less than 1, which can effectively avoid the non-differentiable of $L1$ regularization at point 0.

$$L_1 = \begin{cases} \|x - G(x)\|_1, & x \geq 1; \\ (x - G(x))^2, & x < 1; \end{cases} \quad (2)$$

$$\theta^* = \arg \min_G \max_D V(G, D) + \lambda L_1(G). \quad (3)$$

Equation (3) is the aggregate loss function after adding $L1$ regular term, and λ represents the weight of the regular term, and adjusting λ can affect the generalization ability of the model.

III. GAN METHODS

In this section, we introduce the improvements of our methods in detail, including image and data reprocessing, coding, decoding generator model and discriminator model.

A. Image Preprocessing

Using the generator to randomly generate a low resolution original image will not control the boundary of the generated image, and singular values will appear in some parts of the image. Firstly, the original remote sensing image is reduced to one quarter of the original size, and then stretched to the original size. The image will lose the definition of the image and the details due to compression and stretching. Taking the original image and the scaled image as calibration data pairs, and then training such data pairs to convert the blurred image into a clear image is equivalent to realizing a 4x super-resolution, as shown in Fig. 2. The generator network does not create a low resolution

image from 0, but generates a low resolution image based on reducing and enlarging the image and adding random noise, which can effectively inhibit the possibility of model collapse.

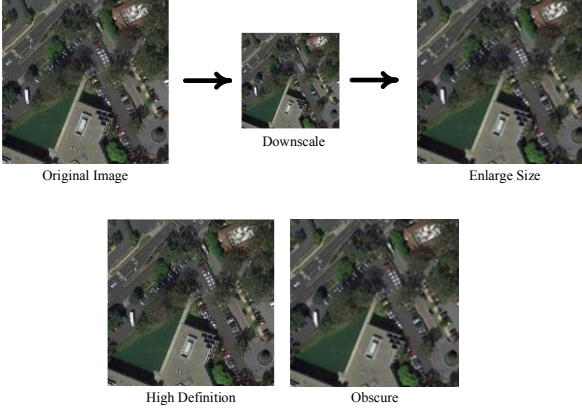


Fig. 2. Blurred image generation

B. Data Preprocessing

The distribution of training data and input data during testing changes too much, which will bring training difficulty to the model. In common multi-layer neural networks, batch regularization is mostly used to restrict the data distribution. Batch regularization is shown as follows:

$$\left. \begin{aligned} \mu_B &\leftarrow \frac{1}{m} \sum_{i=1}^m x_i \\ \sigma_B^2 &\leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_B)^2 \\ \hat{x}_i &\leftarrow \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \\ y_i &\leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \end{aligned} \right\} \quad (4)$$

During training, each batch data input is constrained to mean 0 and variance 1. Similarly, the output is similar to the input. The region represented by the model leads to the deterioration of the generalization ability, so it is necessary to add learnable parameters β and γ Zoom and pan the data. As shown in Fig. 3, input the characteristic map of 128 batches with the size of $32 \times 32 \times 64$, and obtain the characteristic value of 128 batches with the size of $1 \times 1 \times 64$. After 128 batches are fused into a $1 \times 1 \times 1 \times 64$ feature vector. The model will be updated continuously during training β and γ . Batch regularization is to optimize the distribution of input data during batch training of model data.

The generator network task needs to optimize the input data distribution of each image. Therefore, we quotes the idea similar to batch regularization to regularize each feature layer, the formula is as follows:

$$\left. \begin{aligned} \mu_F &\leftarrow \frac{1}{m} \sum_{i=1}^m x_i \\ \sigma_F^2 &\leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_F)^2 \end{aligned} \right\} \quad (5)$$

The principle of Equation (5) is similar to that of Equation (4), except that this statistical feature only occurs in each feature layer and does not count the distribution of batch data.

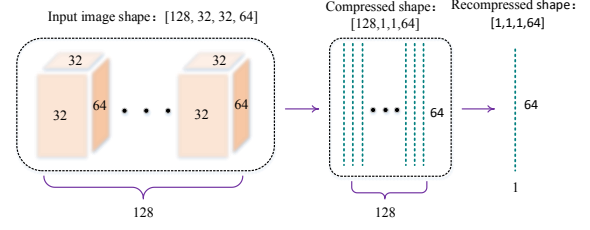


Fig. 3. Batch-Normalization optimization principle

C. Coding and Decoding Generator Model

Dcgan [9] changed the generation network G and discrimination network D into two CNN networks. Based on the idea of encoding and decoding, we designs the process of generating images as the process of compression and decompression of high-definition images. As shown in Fig. 4, a jump connection is added to each layer, and layer i is spliced to layer $n-i$. This is because the image sizes of layer i and layer $n-i$ are the same.

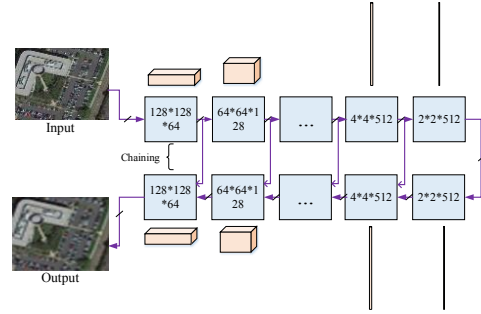


Fig. 4. Encoding and decoding network structure

The input image of the encoding link needs to reduce the image size through convolution operation, then process the batch images through batch regular BATCH NORM [10], regularize a single image through INSTANCE NORM [10] operation, and finally use LeakyReLU [11] to replace the nonlinear activation layer of ReLU [11]. The decoding process is opposite to coding. The last layer needs to use deconvolution upper sampling to restore the original size of the image.

D. Discriminator model

The task of discriminator network is to judge the input image and unknown image. The discriminator in Fig. 5 first splices the two images in the feature dimension, and then enters the convolution layer by layer.

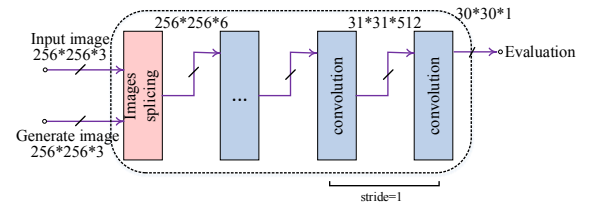


Fig. 5. Generator network structure

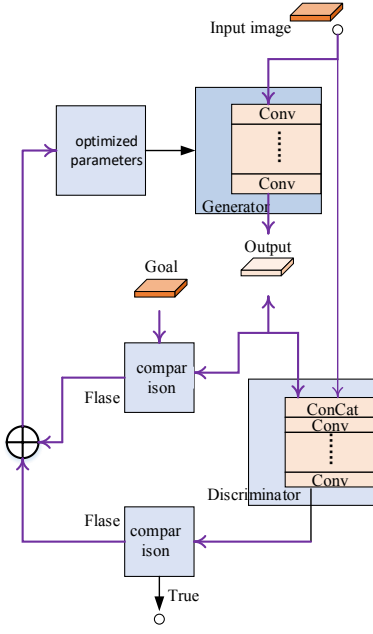


Fig. 6. Generates a confrontation network structure

Instead of using the Softmax function in the original GAN to directly judge true or false, the network uses the eigenvalue with the generated size of 30×30 to calculate the mean value of this grid. As shown in Fig. 6, by selecting the small block of each $N \times N$ in the image to calculate the probability, and then averaging these probabilities as the overall output, the purpose of it is to speed up the calculation speed and accelerate the convergence.

However, it is a compromise between performance and effect in scene that require high generation speed. Of course, you can adjust the value of N to the same size as the original image.

IV. EXPERIMENTAL ANALYSIS AND CONCLUSION

In this section, we use real data sets for experiments to verify the actual effect of this method. At the same time, we conduct comparative experiments with previous methods, and analyze the advantages and disadvantages of this method by comparing the indicators of the generated super-resolution remote sensing image.

A. Experiment

The hardware platform used for network model training is NVIDIA's DGX-1, Intel XeonE5-2698, 8 Tesla P4 GPUs, 256G memory and 4TB SSD. The software platforms are Ubuntu 16.04 operating system, tensorflow 1.8.0-gpu deep learning framework, CUDA9.0 and CUDNN7.14. The datasets adopt DOTA. Peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) are used as objective indicators to measure the quality of image reconstruction. Super resolution image Y and reconstructed image $\tilde{Y}$, PSNR and SSIM are

$$\text{PSNR} = 10 \log \frac{MN}{\|Y - \tilde{Y}\|^2}, \quad (6)$$

$$\text{SSIM} = \frac{(2\mu_Y \mu_{\tilde{Y}} + C_1)(\delta_{Y\tilde{Y}} + C_2)}{(\mu^2_Y + \mu^2_{\tilde{Y}} + C_1)(\delta^2_Y + \delta^2_{\tilde{Y}} + C_2)}. \quad (7)$$

In the above formula, M and N are the size of the image, μ_Y and δ_Y are the average gray value and variance of the original high-resolution image respectively, $\mu_{\tilde{Y}}$ and $\delta_{\tilde{Y}}$ are the gray average value and variance of the reconstructed image respectively, $\delta_{Y\tilde{Y}}$ are the covariance of the original high-resolution image and the reconstructed image, C_1 and C_2 are constants.

B. Comparison and Analysis

It can be seen from table I that the generation of countermeasure network is essentially to recreate a new high-resolution image. Therefore, the objective indicators of traditional PSNR and SSIM are not as good as traditional neural network. Convolutional neural network is really good at super-resolution image reconstruction. In most cases, the index of PSNR is really high, but it does not mean that the display effect is good. Therefore, objective indicators can only be used as a reference and cannot accurately measure the visual effect.

TABLE I. AVERAGING FOUR TIMES LARGER PSNR (DB) AND SSIM

Methods	Indexes	
	PSNR	SSIM
KNN	26.61	0.89
Bictrual Interpolation	28.42	0.81
Bicubic Interpolation	30.28	0.86
SRCNN	30.31	0.86
SRGAN	19.01	0.71
Our method	30.01	0.83

Fig. 7 shows the original remote image in DOTA and the partially enlarged 4x image. Figure (b) uses the original GAN, Figure (c) is the SRCC effect, and Figure (d) is our approach.



Fig. 7. Reconstruction comparison in DOTA datasets

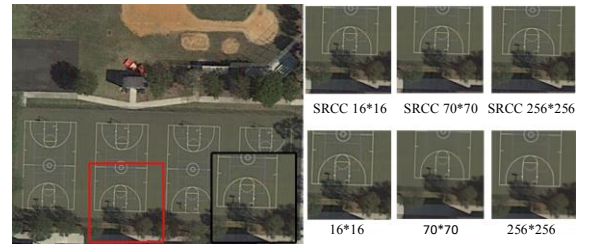


Fig. 8. Different assessment sizes are used in DOTA datasets

It can be seen that although the image generated by the original GAN is similar to the original image in overall style and content, the original GAN is still defective in detail control because there are no restrictions on its generation conditions. The edge of the SRCC is still not clear enough, as shown in Fig.8, but the edge of the SRCC is still not handled well. It can better remove the influence of edge details for the later remote sensing image segmentation and recognition task.

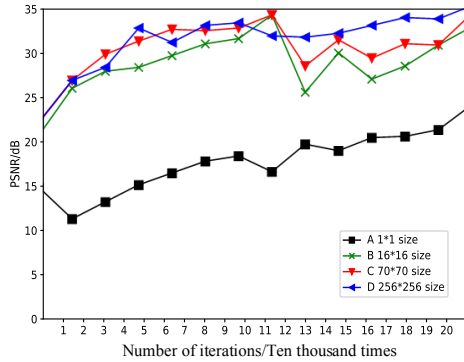


Fig. 9. Block prediction of image

Fig.9 shows the PSNR of high-resolution image reconstruction with different prediction block sizes under different iterations. The smaller the size, the lower the value of PSNR. The larger the size, the higher the PSNR value. After 80000 times, the curve began to stabilize.

TABLE II. COMPARISON OF MODEL PERFORMANCE

size/pix	one step/sec	total time /h	rebuild time/s
1*1	1.1-1.6	about 28	about 1.16
16*16	2.3-3.1	about 30	about 1.26
70*70	3.0-4.2	about 34	about 1.71
256*256	4.1-4.7	about 46	about 1.83

Table II shows the model training time, total training time and image HD reconstruction time under different prediction sizes. For the training phase, the larger the size used, the longer the training time. After the network model and parameters are trained, the time of reconstructing HD is not much different.

V. CONCLUSION

Based on the idea of generating countermeasure network based on deep learning, we improves the deficiency of generator network in SRGAN. The reconstructed remote sensing image quality is improved by using the decompressed model generator instead of GAN. In DOTA training set, the experimental results show that the image generated by this method has advantages in objective quantitative evaluation criteria. In subjective visual evaluation, the reconstructed image has higher edge sharpness, clearer visual effect and natural subjective effect.

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